Detailed Mathematical Formulation of the CAN-7USAT Parafoil Precision-Landing GNC

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Introduction

This document presents the detailed mathematical formulation of the CAN-7USAT parafoil guidance, navigation, trajectory-planning, and control system. A reduced-order wind-aware parafoil model is used for onboard trajectory generation, path following, and brake control.

1 Parafoil Physical Configuration and Aerodynamic Basis

1.1 Physical Configuration

The secondary descent system uses a seven-cell ram-air parafoil. The working geometry is summarized in Table 1.

Table 1: Working parafoil geometry.

Quantity	Symbol	Working value
Flat span	b	$1.00\,\mathrm{m}$
Characteristic chord	c	$\sim 0.50\,\mathrm{m}$
Flat planform area	S	$\sim 0.50\,\mathrm{m}^2$
Flat aspect ratio	AR	~ 2.0
Number of cells	–	7
Number of rib profiles	–	16
Suspended mass	m	$\sim 1.0\,\mathrm{kg}$
Mass loading	m/S	$\sim 2.0\,\mathrm{kg/m}^2$

The ram-air canopy obtains its aerodynamic shape after inflation. Air entering the leading-edge openings establishes internal pressure, while the ribs, fabric tension, suspension lines, and pressure distribution collectively maintain the inflated lifting shape. The canopy therefore behaves as a flexible finite wing rather than as a purely drag-producing parachute.

1.2 Lift and Drag

For a steady air-relative speed V_a, the aerodynamic lift and drag magnitudes are represented by

$$L = \frac{1}{2}\rho V_a^2 S C_L, \tag{1}$$

$$D_a = \frac{1}{2}\rho V_a^2 S C_D, \tag{2}$$

where

- ρ is atmospheric density,
- S is canopy planform area,
- C_L is the lift coefficient,
- C_D is the total drag coefficient,
- D_a denotes aerodynamic drag and is kept distinct from the Down-coordinate used later.

For steady, unaccelerated glide at glide-path angle γ below the horizontal, force balance along and normal to the flight path gives

$$L = mg\cos\gamma, \tag{3}$$

$$D_a = mg\sin\gamma. \tag{4}$$

Dividing Eq. (4) by Eq. (3),

$$\tan\gamma = \frac{D_a}{L} = \frac{C_D}{C_L}. \tag{5}$$

Hence the aerodynamic lift-to-drag ratio is

$$\boxed{G = \frac{L}{D_a} = \frac{C_L}{C_D} = \frac{V_{ah}}{V_D}} \tag{6}$$

where V_{ah} is the horizontal air-relative speed and $V_D > 0$ is the downward air-relative sink speed during nominal glide.

Equation (6) provides the connection between the canopy aerodynamics and the reduced-order GNC quantities used later.

1.3 Working Aerodynamic Coefficients

The values of C_L, C_D, and the Oswald efficiency factor are treated as preliminary design parameters rather than final experimentally identified properties of the fabricated canopy.

Parafoil aerodynamic studies show that leading-edge geometry, canopy flexibility, and trailing-edge deflection materially affect the aerodynamic coefficients, and that the resulting lift and drag values are strongly configuration-dependent [2]. A low-aspect-ratio ram-air wing study uses an Oswald efficiency factor of approximately $e = 0.6$ for a comparable aspect-ratio regime [3]. The following values are therefore adopted as conservative preliminary system-level working values for the expected low-aspect-ratio parafoil glide regime:

$$\boxed{C_L = 0.60, \qquad C_D = 0.25, \qquad e = 0.60.} \tag{7}$$

These values are not direct measurements of the present CAN-7USAT parafoil and are not assumed to be transferable aerodynamic coefficients from any one published configuration. They provide a preliminary coefficient scale for initial performance estimation until the integrated parafoil is characterized through flight testing.

The corresponding nominal glide ratio is

$$G = \frac{0.60}{0.25} = \boxed{2.40}. \tag{8}$$

1.4 Finite-Wing Induced Drag

A common finite-wing approximation is

$$C_D = C_{D0,\text{eff}} + C_{D_i}, \tag{9}$$

with

$$C_{D_i} = \frac{C_L^2}{\pi e AR}. \tag{10}$$

Using $C_L = 0.60$, $e = 0.60$, and $AR = 2.0$,

$$C_{D_i} \approx \frac{0.60^2}{\pi(0.60)(2.0)} \approx 0.095. \tag{11}$$

For the working total $C_D = 0.25$, the remaining effective drag contribution is therefore approximately

$$C_{D0,\text{eff}} \approx 0.25 - 0.095 \approx 0.155. \tag{12}$$

The term $C_{D0,\text{eff}}$ is interpreted broadly. For a ram-air parafoil system it includes losses associated with the flexible canopy, leading-edge openings, fabric shape, suspension lines, rigging, and suspended payload in addition to classical profile drag. The decomposition is therefore used only as an engineering interpretation and not as a high-fidelity aerodynamic identification.

1.5 Nominal Glide Speed Derived from the Working Coefficients

From Eq. (5),

$$\gamma = \tan^{-1}\left(\frac{C_D}{C_L}\right) = \tan^{-1}\left(\frac{0.25}{0.60}\right) \approx 22.6\,\text{deg}. \tag{13}$$

Using Eq. (3),

$$V_a = \sqrt{\frac{2mg\cos\gamma}{\rho S C_L}}. \tag{14}$$

For

$$m = 1.0\,\text{kg}, \quad S = 0.50\,\text{m}^2, \quad \rho = 1.225\,\text{kg/m}^3, \quad C_L = 0.60,$$

the estimated nominal airspeed is approximately

$$V_a \approx 7.0\,\text{m/s}. \tag{15}$$

The corresponding components are

$$V_{ah} = V_a\cos\gamma \approx 6.5\,\text{m/s}, \tag{16}$$

$$V_D = V_a\sin\gamma \approx 2.7\,\text{m/s}. \tag{17}$$

These values provide the preliminary glide-performance estimate used during design. The GNC implementation ultimately uses flight-test-identified V_{ah} and V_D rather than relying only on the aerodynamic prediction.

1.6 Aerodynamic Effect of Steering and Adopted Simplification

A parafoil turns by deflecting one trailing-edge brake line. The resulting asymmetric canopy deformation changes both lift and drag distributions and creates the aerodynamic moments responsible for turning. Parafoil aerodynamic studies show that brake deflection changes both C_L and C_D [2].

For the present reduced-order onboard model, however, the nominal horizontal airspeed and nominal sink rate are treated as independent of steering command:

$$\boxed{V_{ah}(u) \approx V_{ah}, \qquad V_D(u) \approx V_D.} \tag{18}$$

Thus, steering-induced changes in trim speed and glide ratio are not included in the first onboard planning model. This simplification is explicitly revisited during flight-test calibration.

2 Coordinate Systems and Reference Frames

2.1 GNSS Geodetic Coordinate Representation

The NEO-6M receiver outputs navigation data in the NMEA protocol. Latitude and longitude are reported geodetically, and the NMEA GGA message reports altitude above mean sea level together with geoid separation [4]. The quantities used for the local-frame conversion are therefore written as

$$\boxed{(\phi, \lambda, h_{\text{MSL}})} \tag{19}$$

where ϕ is geodetic latitude, λ is geodetic longitude, and h_{MSL} is altitude above mean sea level.

The target reference is represented in the same datum:

$$(\phi_T, \lambda_T, h_{T,\text{MSL}}). \tag{20}$$

NMEA latitude and longitude are first converted from degrees-and-minutes format to decimal degrees and subsequently to radians before the tangent-plane conversion.

2.2 Target-Centred Local Tangent Frame

The fixed ground frame is denoted

$$\boxed{\mathcal{F}_g = \{O_g, x_g, y_g, z_g\}.} \tag{21}$$

Its origin is the designated landing target:

$$O_g \equiv \text{target}. \tag{22}$$

The axes are local North–East–Down:

$$+x_g : \text{North}, \qquad +y_g : \text{East}, \qquad +z_g : \text{Down}. \tag{23}$$

The target therefore has coordinates

$$\boxed{\boldsymbol{p}_T^g = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.} \tag{24}$$

The current CAN-7USAT position in this frame is

$$\boldsymbol{p}^g = \begin{bmatrix} x_g \\ y_g \\ z_g \end{bmatrix}. \tag{25}$$

For the small operational area of the mission, a local tangent-plane approximation is used. Define

$$\Delta\phi = \phi - \phi_T, \tag{26}$$
$$\Delta\lambda = \lambda - \lambda_T. \tag{27}$$

Using the WGS-84 ellipsoid,

$$R_M = \frac{a(1 - e_E^2)}{\left(1 - e_E^2 \sin^2 \phi_T\right)^{3/2}}, \tag{28}$$
$$R_N = \frac{a}{\sqrt{1 - e_E^2 \sin^2 \phi_T}}, \tag{29}$$

where a is the WGS-84 semi-major axis and e_E is the ellipsoid eccentricity.

The target-centred coordinates are approximated by

$$\boxed{x_g = R_M \Delta\phi} \tag{30}$$

$$\boxed{y_g = R_N \cos\phi_T \, \Delta\lambda} \tag{31}$$

$$\boxed{z_g = -\left(h_{\text{MSL}} - h_{T,\text{MSL}}\right)} \tag{32}$$

The target-relative height above the target plane is consequently

$$h = -z_g. \tag{33}$$

2.3 Local Navigation Frame

A moving local navigation frame is defined at the CAN-7USAT centre of mass:

$$\boxed{\mathcal{F}_n = \{O_n, x_n, y_n, z_n\}.} \tag{34}$$

Its origin O_n is the instantaneous CAN-7USAT centre of mass, while its axes remain locally aligned with North, East, and Down:

$$+x_n : \text{North}, \qquad +y_n : \text{East}, \qquad +z_n : \text{Down}. \tag{35}$$

Over the limited operating region, the ground and navigation axes are treated as parallel:

$$\boxed{{}^{g}R_n \approx I_3.} \tag{36}$$

The distinction between $\mathcal{F}_g$ and $\mathcal{F}_n$ is therefore mainly their origin:

- $\mathcal{F}_g$ is fixed at the landing target and is used for path geometry and landing error;
- $\mathcal{F}_n$ translates with the CAN-7USAT and is used for locally level navigation, velocity, wind, and attitude quantities.

2.4 Competition Body Frame

The competition defines three physical positive axes: pitch, roll, and yaw. The positive yaw axis points downward, the positive roll axis points into the page in the competition axis definition, and the positive pitch axis completes the right-handed triad.

For the mathematical formulation, the same physical positive directions are retained, but a cyclic relabelling is used so that the parafoil-forward axis is the first body coordinate:

$$\boxed{\mathcal{F}_b = \{O_b, x_b, y_b, z_b\}} \tag{37}$$

with

$$+x_b : \text{competition pitch axis / parafoil-forward direction}, \tag{38}$$
$$+y_b : \text{competition yaw axis / downward direction}, \tag{39}$$
$$+z_b : \text{competition roll axis / into-page direction}. \tag{40}$$

This ordering is right-handed:

$$\boxed{\hat{\boldsymbol{e}}_{x_b} \times \hat{\boldsymbol{e}}_{y_b} = \hat{\boldsymbol{e}}_{z_b}.} \tag{41}$$

The body-frame origin is the CAN-7USAT centre of mass:

$$O_b = O_n. \tag{42}$$

The adopted parafoil-payload alignment assumption is

$$\boxed{\hat{\boldsymbol{f}}_p \parallel \hat{\boldsymbol{e}}_{x_b},} \tag{43}$$

i.e. the parafoil forward direction is assumed to remain aligned with the CAN-7USAT pitch axis after the parafoil has entered controlled glide.

Let ${}^{n}R_b$ denote the body-to-navigation direction-cosine matrix estimated by the attitude-determination system. The parafoil-forward unit vector expressed in the navigation frame is

$${}^{n}\hat{\boldsymbol{f}}_p = {}^{n}R_b \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} f_N \\ f_E \\ f_D \end{bmatrix}. \tag{44}$$

The geographic parafoil heading is defined as the azimuth of this forward axis projected onto the local horizontal plane:

$$\boxed{\psi = \operatorname{atan2}(f_E, f_N).} \tag{45}$$

Thus,

$$\psi = 0 \quad \text{corresponds to North,} \qquad \psi = \frac{\pi}{2} \quad \text{corresponds to East.}$$

Because the navigation frame is NED, positive ψ is clockwise when viewed from above.

The heading definition in Eq. (45) is used rather than equating geographic heading directly to a particular competition Euler angle.

3 Reduced State Definition

The reduced state used for parafoil guidance and propagation is

$$\boxed{\boldsymbol{x} = \begin{bmatrix} x_g & y_g & z_g & \psi & r \end{bmatrix}^T,} \tag{46}$$

where

- (x_g, y_g, z_g) is the target-relative position,
- ψ is the geographic parafoil heading,
- $r = \dot{\psi}$ is the geographic heading rate.

The symbol r in this reduced model denotes heading rate and is not used as a label for any one competition body axis.

The navigation solution also provides the horizontal ground-velocity components

$$\boldsymbol{V}_{g,h} = \begin{bmatrix} V_{g,x} \\ V_{g,y} \end{bmatrix}. \tag{47}$$

4 Parafoil Kinematics

4.1 Nominal Kinematics

Under the nominal steady-glide approximation, the horizontal air-relative velocity is

$$\boldsymbol{V}_{a,h} = V_{ah} \begin{bmatrix} \cos\psi \\ \sin\psi \end{bmatrix}. \tag{48}$$

Without horizontal wind,

$$\dot{x}_g = V_{ah}\cos\psi, \tag{49}$$
$$\dot{y}_g = V_{ah}\sin\psi, \tag{50}$$
$$\dot{z}_g = V_D. \tag{51}$$

Since $+z_g$ points downward, $V_D > 0$ during descent.

4.2 Kinematics in Horizontal Wind

Define the horizontal wind vector as

$$\boxed{\boldsymbol{W} = \begin{bmatrix} W_x \\ W_y \end{bmatrix},} \tag{52}$$

where $\boldsymbol{W}$ points in the direction in which the air mass moves.

The ground-relative horizontal velocity is the vector sum

$$\boxed{\boldsymbol{V}_{g,h} = \boldsymbol{V}_{a,h} + \boldsymbol{W}.} \tag{53}$$

Therefore,

$$\boxed{\dot{x}_g = V_{ah}\cos\psi + W_x} \tag{54}$$

$$\boxed{\dot{y}_g = V_{ah}\sin\psi + W_y} \tag{55}$$

$$\boxed{\dot{z}_g = V_D} \tag{56}$$

Equations (54)–(56) form the translational part of the reduced onboard model.

5 Reduced Heading Dynamics

A steering command does not produce an instantaneous heading-rate change. The physical response chain is

$$\begin{aligned} &\text{servo motion} \rightarrow \text{brake-line deflection} \rightarrow \text{asymmetric aerodynamic loading} \\ &\qquad \rightarrow \text{turning moment} \rightarrow \text{heading-rate response.} \end{aligned}$$

The aggregate yaw/turn response is represented by a first-order heading-rate model:

$$\boxed{\tau_r \dot{r} + r = K_r u,} \tag{57}$$

where

- $u \in [-1, 1]$ is the normalized steering command,
- K_r is the steady-state heading-rate gain,
- τ_r is the heading-rate time constant.

Therefore,

$$\boxed{\dot{r} = -\frac{1}{\tau_r} r + \frac{K_r}{\tau_r} u.} \tag{58}$$

Since

$$\boxed{\dot{\psi} = r,} \tag{59}$$

the heading is second order with respect to steering input:

$$\tau_r \ddot{\psi} + \dot{\psi} = K_r u. \tag{60}$$

The parameters K_r and τ_r are not taken from literature for the final system. They are identified from steering-response tests of the actual parafoil/CAN-7USAT combination.

5.1 Minimum Turning Radius

Let $R_{\min}$ denote the experimentally identified minimum sustainable turn radius. For an approximately steady turn, the rough consistency relation

$$R_{\min} \approx \frac{V_{ah}}{r_{\max}} \tag{61}$$

can be used, where $r_{\max}$ is the maximum sustainable heading rate.

The planner uses a more conservative radius

$$\boxed{R_{\text{plan}} > R_{\min}} \tag{62}$$

to provide margin for actuator lag, path-following error, wind-model error, and the difference between ideal Dubins curvature transitions and the physical parafoil response.

6 Horizontal-Wind Estimation

6.1 Raw Wind Estimate

The GNSS navigation solution provides ground motion. If horizontal ground speed is V_g and course over ground is χ_g, then

$$V_{g,x} = V_g \cos \chi_g, \tag{63}$$

$$V_{g,y} = V_g \sin \chi_g. \tag{64}$$

From Eq. (53),

$$\boldsymbol{W} = \boldsymbol{V}_{g,h} - \boldsymbol{V}_{a,h}. \tag{65}$$

Thus the instantaneous wind estimate is

$$\boxed{W_{x,\text{raw}} = V_{g,x} - V_{ah} \cos \psi,} \tag{66}$$

$$\boxed{W_{y,\text{raw}} = V_{g,y} - V_{ah} \sin \psi.} \tag{67}$$

6.2 Reason for Conditional Filtering

The attitude sensors are mounted on the suspended CAN-7USAT rather than on the parafoil canopy. Pendulum-like payload motion can therefore cause the measured body orientation to oscillate relative to the mean parafoil flight direction. Since the estimated air-relative velocity direction in Eqs. (66)–(67) depends directly on ψ, such oscillation can be incorrectly interpreted as wind variation.

The raw wind estimate is consequently accepted only under valid wind-estimation conditions. Define

$$\gamma_k = \begin{cases} 1, & \text{wind-estimation sample accepted,} \\ 0, & \text{wind-estimation sample rejected.} \end{cases} \tag{68}$$

The filtered estimate is updated using an exponential moving average:

$$\boxed{\hat{\boldsymbol{W}}_k = \begin{cases} \alpha \boldsymbol{W}_{\text{raw},k} + (1-\alpha)\hat{\boldsymbol{W}}_{k-1}, & \gamma_k = 1, \\ \hat{\boldsymbol{W}}_{k-1}, & \gamma_k = 0. \end{cases}} \tag{69}$$

The validity logic rejects samples associated with significant turning, steering transients, and payload attitude oscillation. The numerical gates and α are obtained through simulation and flight-test tuning.

7 Guidance and Trajectory Planning

7.1 Landing Objective

Because the ground frame is centred at the designated target,

$$\boldsymbol{p}_T^g = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \tag{70}$$

The horizontal landing error is

$$\boxed{e_L = \sqrt{x_g^2 + y_g^2}.} \tag{71}$$

The trajectory planner must reduce e_L while respecting the available descent height, minimum-turn-radius constraint, wind, and terminal into-wind approach.

7.2 Final-Approach Heading

Let the filtered wind estimate be

$$\hat{\boldsymbol{W}} = \begin{bmatrix} \hat{W}_x \\ \hat{W}_y \end{bmatrix}. \tag{72}$$

Its direction of motion is

$$\psi_{W,\mathrm{to}} = \mathrm{atan2}(\hat{W}_y, \hat{W}_x). \tag{73}$$

The desired final heading is opposite the wind-motion direction:

$$\boxed{\psi_f = \mathrm{wrap}\left(\psi_{W,\mathrm{to}} + \pi\right).} \tag{74}$$

Because the final approach is aligned directly opposite the estimated wind direction, its nominal crosswind component is zero. The final ground-track direction is therefore

$$\boxed{\chi_f = \psi_f.} \tag{75}$$

The wrap operator is defined by

$$\mathrm{wrap}(\theta) = \mathrm{atan2}(\sin\theta, \cos\theta). \tag{76}$$

7.3 Final-Approach Entry Point

Directly steering toward the target does not reserve a controlled terminal segment and does not guarantee the desired landing direction. A final approach of prescribed horizontal length L_f is therefore reserved.

The approach-entry point is located L_f behind the target along the desired final heading:

$$\boxed{x_A = -L_f \cos\chi_f,} \tag{77}$$

$$\boxed{y_A = -L_f \sin\chi_f.} \tag{78}$$

The final-approach configuration is

$$\boxed{q_A = (x_A, y_A, \chi_f).} \tag{79}$$

The nominal guidance problem is therefore divided into

$$\text{current configuration} \rightarrow q_A \rightarrow \text{target}.$$

The second portion is a straight final segment of length L_f aligned with χ_f.

7.4 Classical Dubins Path

7.4.1 Ground-Track Configuration

The Dubins geometry is defined in the ground plane. The path-orientation variable is therefore the ground-track tangent angle χ, not the air-relative parafoil heading ψ.

The current ground-track angle is obtained from the horizontal GNSS ground-velocity components:

$$\boxed{\chi_0 = \operatorname{atan2}(V_{g,y}, V_{g,x}).} \tag{80}$$

The initial planar configuration is

$$\boxed{q_0 = (x_g, y_g, \chi_0).} \tag{81}$$

The final Dubins configuration is the final-approach entry configuration $q_A = (x_A, y_A, \chi_f)$.

7.4.2 Dubins Vehicle Model

A classical Dubins vehicle is a forward-moving planar vehicle with bounded curvature. Parameterizing the ground path by arc length s gives

$$\frac{dx_g}{ds} = \cos\chi, \tag{82}$$

$$\frac{dy_g}{ds} = \sin\chi, \tag{83}$$

$$\frac{d\chi}{ds} = \kappa, \tag{84}$$

where the curvature satisfies

$$\boxed{|\kappa| \le \frac{1}{R_{\text{plan}}}.} \tag{85}$$

Thus, a shortest Dubins path is composed only of

- a maximum-curvature clockwise/right arc,
- a straight segment,
- a maximum-curvature counter-clockwise/left arc.

With the present North–East ground-plane convention, increasing χ corresponds to a clockwise/right turn when viewed from above.

7.4.3 Left- and Right-Turn Circles

For a planar configuration

$$q = (x, y, \chi) \tag{86}$$

and planning radius R_{plan}, the right-turn circle centre is

$$\boxed{\boldsymbol{C}_R(q) = \begin{bmatrix} x - R_{\text{plan}}\sin\chi \\ y + R_{\text{plan}}\cos\chi \end{bmatrix},} \tag{87}$$

while the left-turn circle centre is

$$\boxed{\boldsymbol{C}_L(q) = \begin{bmatrix} x + R_{\text{plan}}\sin\chi \\ y - R_{\text{plan}}\cos\chi \end{bmatrix}.} \tag{88}$$

At $\chi = 0$, the path points North. The right-turn circle lies to the East and the left-turn circle lies to the West, consistent with the adopted clockwise-positive ground-track convention.

7.4.4 Six Classical Path Families

The shortest Dubins path between an initial configuration q_0 and a final configuration q_f belongs to one of six families:

$$\boxed{LSL, \quad RSR, \quad LSR, \quad RSL, \quad RLR, \quad LRL.} \tag{89}$$

The first four are curve–straight–curve (CSC) paths. Their straight segments correspond to appropriate common tangents between the initial and final turn circles.

The final two are curve–curve–curve (CCC) paths. They become relevant when the configurations are sufficiently close that a three-arc solution is shorter than the available CSC alternatives.

For each feasible family i, the geometric path length is

$$L_i = R_{\text{plan}}\Delta\chi_{1,i} + L_{S,i} + R_{\text{plan}}\Delta\chi_{2,i} \tag{90}$$

for a CSC path, or

$$L_i = R_{\text{plan}}\left(\Delta\chi_{1,i} + \Delta\chi_{2,i} + \Delta\chi_{3,i}\right) \tag{91}$$

for a CCC path.

A conventional Dubins planner selects

$$\boxed{P^{\star}_{\text{Dubins}} = \arg\min_i L_i.} \tag{92}$$

7.5 Why the Classical Shortest Dubins Path Is Insufficient

Equation (92) is purely geometric. It does not consider the fact that the parafoil is unpowered and continuously loses height.

For the CAN-7USAT, two geometrically valid paths can have very different flight consequences:

- a path that is too short can bring the vehicle horizontally to the terminal region while substantial height remains;
- a path that is too long can require more flight time than the available descent height allows;
- wind changes the traversal time of different ground-track directions even if their geometric lengths are identical.

Wind-aware parafoil trajectory-planning literature similarly treats horizontal geometry, wind, and descent-height consumption as coupled quantities [1]. The adopted planner therefore retains the closed-form Dubins geometry but augments it with wind-aware traversal-time prediction and altitude-feasibility evaluation:

$$\boxed{\text{Dubins ground-path geometry} + \text{wind-aware propagation} + \text{altitude feasibility.}}$$

The nominal path from the current configuration to the target is constructed as

$$\boxed{P_i = P_{D,i}(q_0, q_A) \oplus P_f(q_A, \boldsymbol{p}_T),} \tag{93}$$

where $P_{D,i}$ is one Dubins candidate to the approach-entry configuration and P_f is the reserved straight final approach.

The planner is evaluated only during initial path generation and accepted replanning events. The inner heading-control loop does not repeat the full candidate-path propagation.

7.6 Energy-Management Point

Deliberate height-loss manoeuvres are not performed early in the descent. Altitude loss is irreversible, and preserving excess height is safer than prematurely consuming height based on an uncertain early wind or glide estimate.

For each Dubins connection $P_{D,i}$, parameterized by ground-path arc length s,

$$P_{D,i}(s), \qquad 0 \le s \le L_{D,i}, \tag{94}$$

an energy-management point is defined a prescribed path distance L_E upstream of the final-approach entry point:

$$\boxed{q_{E,i} = P_{D,i}(L_{D,i} - L_E).} \tag{95}$$

The selected path therefore contains the ordered terminal sequence

$$\boxed{q_0 \to q_E \to q_A \to \boldsymbol{p}_T.}$$

The vehicle follows the selected path toward q_E without deliberate height-loss loitering. When the path geometry permits, q_E is placed on a straight portion of the selected path so that the most recent accepted straight-flight wind estimate is available before the energy-management decision.

The design distances L_E and L_f are determined experimentally. They are chosen large enough to provide room for loiter exit, path reconnection, and stable final approach, while remaining sufficiently close to the target that the reserved terminal geometry does not unnecessarily reduce overall reachability.

7.7 Wind-Aware Propagation of a Candidate Ground Path

A Dubins curve specifies a desired *ground path*. In wind, the parafoil heading ψ is generally not equal to the ground-track tangent angle χ because the parafoil must crab into the crosswind.

Consider a discretized ground-path element j of length Δs_j and ground-track tangent angle χ_j. Define

$$\boldsymbol{t}_j = \begin{bmatrix} \cos\chi_j \\ \sin\chi_j \end{bmatrix}, \tag{96}$$

$$\boldsymbol{n}_j = \begin{bmatrix} -\sin\chi_j \\ \cos\chi_j \end{bmatrix}. \tag{97}$$

Resolve the filtered wind into tangent and normal components:

$$W_{\parallel,j} = \hat{\boldsymbol{W}} \cdot \boldsymbol{t}_j, \tag{98}$$

$$W_{\perp,j} = \hat{\boldsymbol{W}} \cdot \boldsymbol{n}_j. \tag{99}$$

To remain on the desired ground path, the normal component of the air-relative velocity must cancel the crosswind:

$$V_{ah}\sin(\psi_j - \chi_j) + W_{\perp,j} = 0. \tag{100}$$

Therefore the required parafoil heading is

$$\boxed{\psi_j = \text{wrap}\left[\chi_j - \sin^{-1}\left(\frac{W_{\perp,j}}{V_{ah}}\right)\right].} \tag{101}$$

A necessary directional-feasibility condition is

$$\boxed{|W_{\perp,j}| \le V_{ah}.} \tag{102}$$

The airspeed remaining in the along-track direction is

$$V_{a,\parallel,j} = \sqrt{V_{ah}^2 - W_{\perp,j}^2}. \tag{103}$$

The resulting along-track ground speed is

$$\boxed{V_{g,j} = W_{\parallel,j} + \sqrt{V_{ah}^2 - W_{\perp,j}^2}.} \tag{104}$$

The path element is usable only if

$$\boxed{V_{g,j} > 0.} \tag{105}$$

No element-by-element heading-rate feasibility test is performed onboard. Instead, the experimentally identified $R_{\min}$ is replaced in planning by the more conservative R_{plan}, and the adequacy of this margin is validated offline using the identified heading dynamics. This avoids adding a dynamic feasibility calculation to every discretized path element.

Its traversal time is

$$\boxed{\Delta t_j = \frac{\Delta s_j}{V_{g,j}}.} \tag{106}$$

For a candidate path containing M_i elements,

$$\boxed{T_i = \sum_{j=1}^{M_i} \Delta t_j.} \tag{107}$$

Thus, wind enters the planner through the time required to execute the geometric path.

7.8 Altitude Propagation

Under the constant-sink approximation of Eq. (18), the candidate terminal Down coordinate is

$$\boxed{z_{f,i} = z_0 + V_D T_i.} \tag{108}$$

Because the target plane is $z_T = 0$, define the candidate vertical endpoint error as

$$\boxed{e_{z,i} = z_T - z_{f,i} = -z_{f,i}.} \tag{109}$$

The sign of $e_{z,i}$ has a direct interpretation:

- $e_{z,i} > 0$: the candidate reaches the horizontal terminal condition while the CAN-7USAT is still above the target plane;
- $e_{z,i} = 0$: the candidate consumes the available height exactly in the reduced model;
- $e_{z,i} < 0$: the candidate requires more descent depth than is available and therefore reaches the target plane before the path is completed.

7.9 Initial Remaining-Flight Situation Evaluation

Let $z_{\text{tol}} > 0$ denote the altitude-feasibility tolerance. For every wind-feasible candidate path, the predicted terminal Down-coordinate and vertical endpoint error are obtained from Eqs. (108)–(109).

The candidates are divided into

$$\mathcal{P}_{\text{nom}} = \{P_i : |e_{z,i}| \le z_{\text{tol}}\}, \tag{110}$$

$$\mathcal{P}_{\text{excess}} = \{P_i : e_{z,i} > z_{\text{tol}}\}, \tag{111}$$

$$\mathcal{P}_{\text{insuff}} = \{P_i : e_{z,i} < -z_{\text{tol}}\}. \tag{112}$$

The classification is based on wind-aware traversal time rather than geometric path length alone. An excess-height classification does not trigger immediate loitering; deliberate height dissipation is deferred until the selected path reaches q_E.

7.10 Turning-Effort Metric and Initial Path Selection

The altitude condition is treated as a feasibility requirement rather than as a weighted term in an optimization cost. For candidate i, define the total planned Dubins turning angle

$$\boxed{\Theta_i = \sum_k \left|\Delta\chi_{k,i}\right|.} \tag{113}$$

If one or more nominal-height candidates exist, the selected nominal path is

$$\boxed{P^\star = \arg\min_{P_i \in \mathcal{P}_{\text{nom}}} \Theta_i.} \tag{114}$$

If no nominal candidate exists but one or more excess-height candidates exist, the selected path is

$$\boxed{P^\star = \arg\min_{P_i \in \mathcal{P}_{\text{excess}}} \Theta_i.} \tag{115}$$

Minimizing Θ_i reduces unnecessary steering, brake activity, and dependence on uncertain turn dynamics without introducing an iterative nonlinear optimizer.

If all wind-feasible candidates lie in $\mathcal{P}_{\text{insuff}}$, the terminal-entry constraint is abandoned and the insufficient-height logic defined below is used.

7.11 Energy Evaluation at the Energy-Management Point

When the selected path reaches q_E, the remaining path

$$q_E \to q_A \to \boldsymbol{p}_T$$

is propagated again using the current navigation state and the latest accepted horizontal-wind estimate.

Let

$$T_{E\to T} \tag{116}$$

be the predicted traversal time of this remaining path. The available flight time before reaching the target plane is

$$\boxed{T_{\text{avail},E} = \frac{-z_E}{V_D}.} \tag{117}$$

The predicted excess flight time is

$$\boxed{\Delta T_E = T_{\text{avail},E} - T_{E\to T}.} \tag{118}$$

Equivalently, define the predicted vertical margin at q_E as

$$\boxed{e_{z,E} = V_D \Delta T_E.} \tag{119}$$

A positive $e_{z,E}$ means that height remains after completing the nominal remaining path; a negative value means the nominal remaining path requires more height than is available.

A positive exit reserve $z_{\text{res}} > 0$ is retained so that the energy-management manoeuvre is biased toward leaving the vehicle slightly high rather than making the target unreachable.

7.12 Nominal-Height Case at q_E

If

$$-z_{\text{tol}} \le e_{z,E} \le z_{\text{res}}, \tag{120}$$

no deliberate height-loss manoeuvre is performed. The vehicle proceeds directly along the remaining path:

$$\boxed{q_E \to q_A \to \boldsymbol{p}_T.}$$

7.13 Insufficient-Height Case

If

$$\boxed{e_{z,E} < -z_{\text{tol}},} \tag{121}$$

the prescribed final-approach entry configuration is no longer retained as an objective. The guidance mode changes to best-effort direct-to-target tracking.

The desired ground-track angle is

$$\boxed{\chi_{\text{BE}} = \text{atan2}(-y_g, -x_g).} \tag{122}$$

The corresponding parafoil heading is obtained using the same wind-triangle correction used by the normal LOS guidance. Thus, when the complete terminal geometry is no longer reachable, the onboard objective becomes simply to reduce the horizontal touchdown miss distance as much as possible.

7.14 Excess-Height Management at q_E

If

$$\boxed{e_{z,E} > z_{\text{res}},} \tag{123}$$

the vehicle enters a constant-radius loiter manoeuvre tangent to the selected path at q_E.

Let

$$q_E = (x_E, y_E, \chi_E). \tag{124}$$

A right-hand tangent loiter circle of radius $R_L \geq R_{\text{plan}}$ has centre

$$\boldsymbol{C}_{E,R} = \begin{bmatrix} x_E - R_L \sin\chi_E \\ y_E + R_L \cos\chi_E \end{bmatrix}, \tag{125}$$

while the left-hand alternative has centre

$$\boldsymbol{C}_{E,L} = \begin{bmatrix} x_E + R_L \sin\chi_E \\ y_E - R_L \cos\chi_E \end{bmatrix}. \tag{126}$$

Either tangent direction may be used; the selected direction is determined from the available terminal geometry and does not change the following height-management calculation.

One complete orbit returns the vehicle to the same ground position and ground-track direction q_E. Consequently, the remaining path $q_E \to q_A$ remains geometrically valid after every complete orbit.

If one orbit is divided into M_L elements, its wind-aware traversal time is

$$\boxed{T_{\text{orbit}} = \sum_{j=1}^{M_L} \frac{\Delta s_j}{V_{g,j}}.} \tag{127}$$

The Down-coordinate consumed by one complete orbit is

$$\boxed{\Delta z_{\text{orbit}} = V_D T_{\text{orbit}}.} \tag{128}$$

A full orbit is executed only while the predicted excess height is large enough that the reserved terminal path remains reachable after the orbit. After each complete orbit, the remaining-height condition is reevaluated using the current navigation state and the latest accepted wind estimate. Because wind-estimation samples are rejected during turning, the wind estimate itself is updated only when valid approximately straight-flight samples become available.

If another complete orbit would consume more than the required excess above the reserve, only the residual height

$$\Delta z_{\text{frac}} = e_{z,E} - z_{\text{res}} \tag{129}$$

is intentionally dissipated.

The corresponding partial orbit is generated by advancing through the discretized loiter circle until the accumulated predicted sink satisfies

$$\boxed{V_D \sum_{j=1}^{M_{\text{frac}}} \Delta t_j \approx \Delta z_{\text{frac}}.} \tag{130}$$

The partial-orbit exit configuration is

$$q_{E'} = (x_{E'}, y_{E'}, \chi_{E'}). \tag{131}$$

One new Dubins connection is then generated from $q_{E'}$ to q_A:

$$\boxed{P_{D,\text{new}} : q_{E'} \to q_A.} \tag{132}$$

The vehicle subsequently follows

$$\boxed{q_{E'} \to q_A \to \boldsymbol{p}_T.}$$

This late energy-management strategy deliberately preserves height during the earlier part of the descent and dissipates only confirmed excess height near the terminal region.

7.15 Line-of-Sight Path Following

The planner generates a geometric reference path. A line-of-sight (LOS) guidance law converts this path into a continuously updated desired heading.

Let

$$\boldsymbol{p}_{\text{LOS}} = \begin{bmatrix} x_{\text{LOS}} \\ y_{\text{LOS}} \end{bmatrix} \tag{133}$$

be a point located a prescribed look-ahead distance L_{LOS} ahead of the closest/reference point on the active path.

The desired ground-track direction is

$$\boxed{\chi_d = \operatorname{atan2}\left(y_{\text{LOS}} - y_g, x_{\text{LOS}} - x_g\right).} \tag{134}$$

Because the vehicle must crab into wind, the required parafoil heading is computed from the same wind triangle used during trajectory propagation.

Define

$$\boldsymbol{t}_d = \begin{bmatrix} \cos\chi_d \\ \sin\chi_d \end{bmatrix}, \tag{135}$$

$$\boldsymbol{n}_d = \begin{bmatrix} -\sin\chi_d \\ \cos\chi_d \end{bmatrix}, \tag{136}$$

$$W_{\perp,d} = \hat{\boldsymbol{W}} \cdot \boldsymbol{n}_d. \tag{137}$$

The desired heading command is

$$\boxed{\psi_d = \text{wrap}\left[\chi_d - \sin^{-1}\left(\frac{W_{\perp,d}}{V_{ah}}\right)\right].} \tag{138}$$

The wind correction is evaluated only when

$$\boxed{|W_{\perp,d}| < V_{ah}.} \tag{139}$$

Violation of Eq. (139) means that the requested LOS ground track cannot be maintained with the available horizontal airspeed and therefore requests replanning. No additional heading-rate feasibility optimization is performed inside the LOS guidance loop.

This separates path geometry from wind compensation:

$$\text{reference path} \rightarrow \chi_d \rightarrow \text{wind correction} \rightarrow \psi_d.$$

7.16 Replanning Logic

The active path is generated using the estimated state, estimated wind, and identified nominal parafoil performance. Replanning is therefore required when the active path no longer provides an adequate prediction of the remaining flight.

Three replanning quantities are monitored.

7.16.1 Cross-Track Error

For an active straight segment with endpoints

$$\boldsymbol{p}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \qquad \boldsymbol{p}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}, \tag{140}$$

define

$$\boldsymbol{s} = \boldsymbol{p}_2 - \boldsymbol{p}_1, \tag{141}$$

$$\boldsymbol{r}_p = \begin{bmatrix} x_g \\ y_g \end{bmatrix} - \boldsymbol{p}_1. \tag{142}$$

The signed straight-segment cross-track error is

$$\boxed{e_{\text{xt}} = \frac{s_x r_{p,y} - s_y r_{p,x}}{\|\boldsymbol{s}\|_2}.} \tag{143}$$

For an active circular arc of centre

$$\boldsymbol{C} = \begin{bmatrix} C_x \\ C_y \end{bmatrix} \tag{144}$$

and radius R, the cross-track magnitude is

$$\boxed{|e_{\text{xt}}| = \left| \sqrt{(x_g - C_x)^2 + (y_g - C_y)^2} - R \right|.} \tag{145}$$

A path-deviation trigger is

$$\boxed{|e_{\text{xt}}| > e_{\text{xt,max}}.} \tag{146}$$

7.16.2 Wind-Estimate Change

The wind estimate continues to update whenever accepted approximately straight-flight samples are available. Let $\boldsymbol{W}_{\text{plan}}$ denote the filtered wind estimate used when the currently active path was generated. A wind-change trigger is

$$\boxed{\left\| \hat{\boldsymbol{W}} - \boldsymbol{W}_{\text{plan}} \right\|_2 > \Delta W_{\text{replan}}.} \tag{147}$$

7.16.3 Remaining-Altitude Consistency

At the current state, let T_{rem} be the newly predicted wind-aware traversal time of the remaining active path. The predicted terminal Down-coordinate is

$$z_{f,\text{pred}} = z_g + V_D T_{\text{rem}}. \tag{148}$$

Define

$$\boxed{e_{z,\text{rem}} = z_T - z_{f,\text{pred}}.} \tag{149}$$

An altitude-consistency trigger is

$$\boxed{|e_{z,\text{rem}}| > e_{z,\text{replan}}.} \tag{150}$$

When a replanning trigger is accepted, the guidance problem is reconstructed from the current estimated state and current filtered wind estimate.

8 Terminal Flare Mode

The flare is treated as a terminal mode distinct from normal path-following guidance. When the configured flare-initiation condition is asserted, normal heading tracking is terminated and a symmetric brake command is applied:

$$\boxed{\delta_L = \delta_R = \delta_f.} \tag{151}$$

The flare command δ_f and the flare-entry condition are determined through descent-system testing.

9 Heading Control

9.1 Heading Error

The desired heading from LOS guidance is ψ_d. The wrapped heading error is

$$\boxed{e_\psi = \text{wrap}(\psi_d - \psi) = \text{atan2}\left[\sin(\psi_d - \psi), \cos(\psi_d - \psi)\right].} \tag{152}$$

The wrap operation ensures that the commanded correction always corresponds to the shortest angular difference.

9.2 PD Heading Controller

The unsaturated steering command is

$$\boxed{u^\star = K_p e_\psi - K_d r.} \tag{153}$$

The proportional term generates the turning demand and the heading-rate term provides damping.

The command is saturated:

$$\boxed{u_c = \text{clip}(u^\star, -1, 1).} \tag{154}$$

A command-rate limit is then applied:

$$\boxed{u_k = u_{k-1} + \text{clip}\left(u_{c,k} - u_{k-1}, -\dot{u}_{\max}\Delta t, +\dot{u}_{\max}\Delta t\right).} \tag{155}$$

9.3 Single-Sided Brake Allocation

Normal steering uses one brake at a time. Define normalized left and right brake demands

$$\delta_L, \delta_R \in [0, 1]. \tag{156}$$

The allocation law is

$$\boxed{\delta_L = \max(-u, 0),} \tag{157}$$

$$\boxed{\delta_R = \max(u, 0).} \tag{158}$$

Thus,

$$u > 0 \quad \Rightarrow \quad \delta_R = u,\ \delta_L = 0,$$

and

$$u < 0 \quad \Rightarrow \quad \delta_L = -u,\ \delta_R = 0.$$

The normalized brake demands are converted to servo commands using the experimentally calibrated servo-angle/PWM mapping.

10 Compact Final Model

The reduced-order propagation model used by the GNC formulation is

$$\boxed{\begin{aligned} \dot{x}_g &= V_{ah}\cos\psi + \hat{W}_x, \\ \dot{y}_g &= V_{ah}\sin\psi + \hat{W}_y, \\ \dot{z}_g &= V_D, \\ \dot{\psi} &= r, \\ \dot{r} &= -\frac{1}{\tau_r} r + \frac{K_r}{\tau_r} u. \end{aligned}} \tag{159}$$

The primary constraints are

$$\boxed{|u| \le 1, \qquad |\dot{u}| \le \dot{u}_{\max}, \qquad R_{\text{path}} \ge R_{\text{plan}} > R_{\min}.} \tag{160}$$

The onboard information flow is summarized as

$$\boxed{\begin{aligned} \text{GNSS + attitude solution} &\rightarrow \text{target-centred navigation state} \\ &\rightarrow \text{wind estimation} \\ &\rightarrow \text{wind-aware trajectory planning} \\ &\rightarrow \text{LOS guidance} \\ &\rightarrow \text{heading control} \\ &\rightarrow \text{single-sided brake actuation.} \end{aligned}} \tag{161}$$

11 Model Assumptions

The reduced formulation uses the following assumptions:

1. The parafoil is fully deployed and in controlled glide before the GNC model is applied.
2. The parafoil forward direction is aligned with the CAN-7USAT competition pitch axis in the mean, as stated in Eq. (43).

3. The target-centred tangent frame and the moving navigation frame are treated as parallel over the mission operating region.

4. The nominal horizontal airspeed V_{ah} and sink speed V_D are constant model parameters during ordinary glide.

5. Steering-induced changes in trim speed, sink rate, and glide ratio are not included in the first onboard planner.

6. The wind used by the planner is represented by its horizontal components (W_x, W_y).

7. Dubins geometry is applied in the ground plane using the conservative radius R_{plan}.

8. The difference between ideal instantaneous Dubins curvature transitions and the physical turn transient is handled through the conservative planning radius, LOS tracking, command-rate limiting, and replanning rather than an element-by-element dynamic feasibility test onboard.

9. The aerodynamic design coefficients in Eq. (7) are preliminary values; the operational GNC quantities are corrected through flight-test identification.

12 Parameters to Be Identified, Calibrated, or Tuned

12.1 Physical/Aerodynamic Quantities

The following quantities are known from design geometry or are retained as preliminary aerodynamic values:

Parameter	Meaning	Current status
b	Parafoil span	Design geometry
c	Characteristic chord	Design geometry
S	Planform area	Design geometry
AR	Aspect ratio	Derived from geometry
C_L	Nominal lift coefficient	Preliminary literature-supported value
C_D	Nominal drag coefficient	Preliminary literature-supported value
e	Oswald efficiency factor	Preliminary literature-supported value

12.2 Flight-Test-Identified Parafoil Parameters

The following parameters are obtained from glide and steering tests of the actual integrated system:

$$\boxed{V_{ah}, \quad V_D, \quad K_r, \quad \tau_r, \quad R_{\min}.} \tag{162}$$

Useful supporting identified quantities include

$$r_{\max} \tag{163}$$

and the measured steady-turn radius as a function of brake command.

12.3 Guidance and Control Tuning Parameters

The principal guidance and control parameters are

$$\boxed{R_{\text{plan}},\quad R_L,\quad L_E,\quad L_f,\quad L_{\text{LOS}},\quad \alpha,\quad K_p,\quad K_d,\quad \dot{u}_{\max}.} \tag{164}$$

The replanning thresholds are

$$\boxed{e_{\text{xt,max}},\qquad \Delta W_{\text{replan}},\qquad e_{z,\text{replan}},\qquad z_{\text{tol}},\qquad z_{\text{res}}.} \tag{165}$$

The terminal-control calibration includes

$$\boxed{\delta_f} \tag{166}$$

and the left/right servo command mapping.

These values are finalized through simulation, hardware-in-the-loop testing, and parafoil drop-test characterization.

13 References

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